Programming Paradigms

CSI 2120

Programming Paradigms

- Paradigm:

  "A philosophical and theoretical framework of a scientific school or discipline within which theories, laws, and generalizations and the experiments performed in support of them are formulated" (Merriam-Webster).

  .

- Programming:

  "The process or activity of writing computer programs."

Programming Paradigms

- Programming Langage:
  A programming language is an artificial language designed to communicate instructions to a computer to perform specific tasks. A language is generally dominated by a paradigm from which its structure was designed.
- To program:
  To use a programming language following a programming paradigm in a more or less rigorous way in order to create a computer program which can be executed by the computer to achieve a desired outcome.

Programming Paradigms

No single paradigm is suitable for all applications.

Any problem can be solved by relying on one or another programming paradigm

No programming languages purely implement a given paradigm

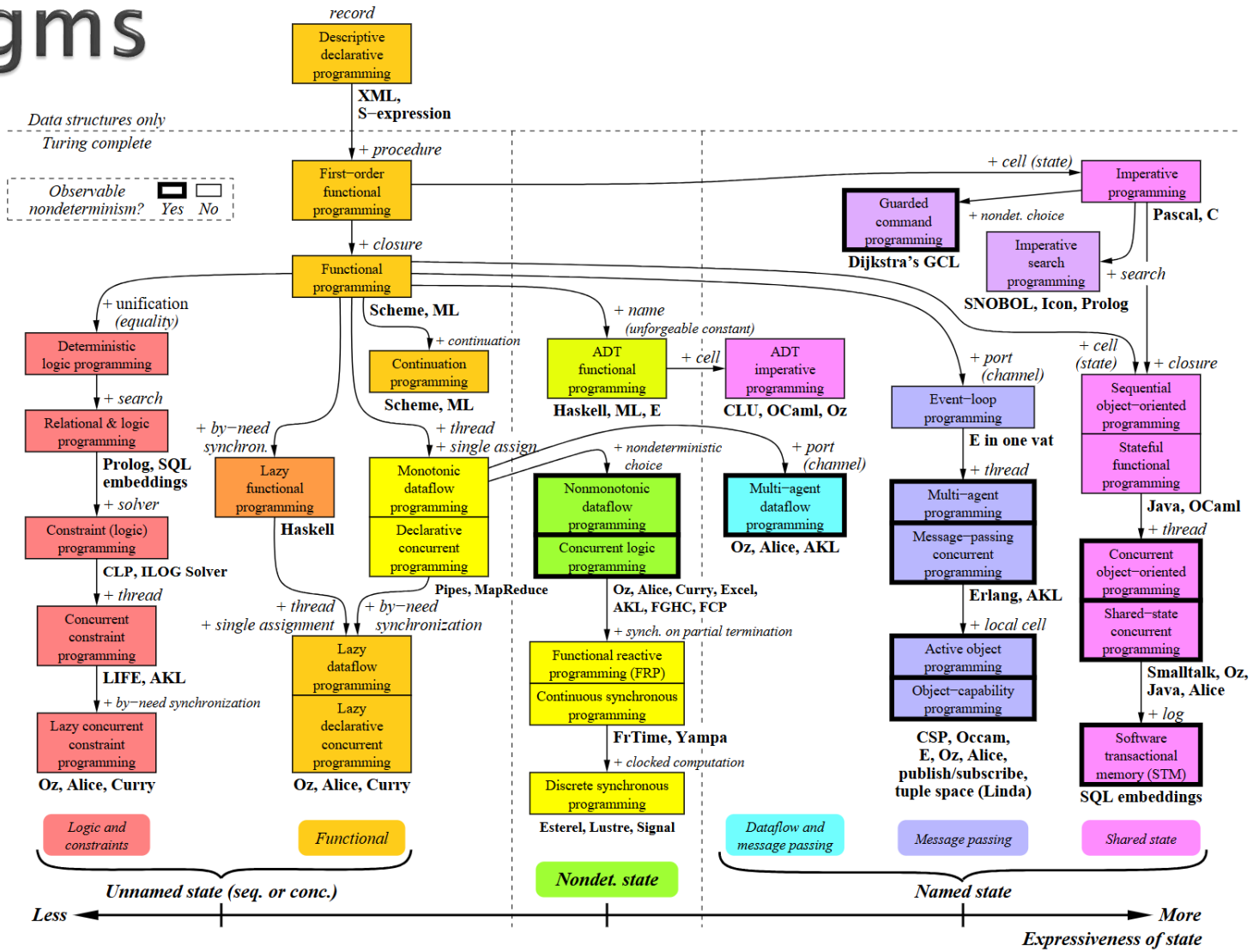
All languages facilitate programming in more than one paradigm

We will study here:

1. Object-oriented paradigm
   - with Java
2. Imperative (procedural) paradigm
   - with Go
3. Concurrent programming pradigm
   - with Go
4. Logic programming paradigm
   - with Prolog
5. Functional programming paradigm
   - with Scheme

But there's many more…

Taxonomy of Programming Paradigms



Why studying different programming paradigms?

- To better express complex ideas
- To solve problems using the most appropriate solution
- To facilitate the learning of new programming languages
- To make informed and effective choices
- To keep up with the evolution of the state of the art in programming
- To become better programmers